

The Layout of Web Pages: A Study on the Relation between Information Forms and Locations Using Eye-Tracking

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Abstract. Web pages are an important human-computer interface, and the layout of Web pages has an effect on visual search. This study focuses on using eye-tracking to explore the relation between information forms and locations, as well as the impact of floating advertisements (ads for short) for visual search on Web pages. The results show that for the information presented as text, the fixation duration in quadrant I (right upper), II (left upper) and III (left lower) are shorter than in quadrant IV(right lower) and the center area; for the picture, the fixation duration in quadrant II is the shortest and in quadrant III is the longest. No matter what location the information is placed on a Web page, there is no significant difference between the Web page with floating ads and without floating ads on the fixation duration, which indicates that floating ads have no significant effect on the layout of Web pages. The results suggest that the important information whether text or a picture is suitable to place in quadrant II; however, it is the worst for the layout of Web pages to arrange text information in quadrant IV and the center area or picture information in quadrant III.

1 Introduction

The Internet is playing an important role in people's daily life, and becoming one of the main channels from which people can get their desired information. Web pages are the carrier of Internet information and the important human-computer interface. The users' visual search efficiency is directly related to the quality of Web page design, thus it is hard to find the desired information quickly on an irrational Web page design. Users' visual search on Web pages is impacted by many factors, such as colors [1,2], font size [3], information forms [4,5] and the scan path [6]. And the information layout of Web pages is an important aspect in Web page design.

The users' visual behavior on Web pages includes two types which are visual browsing without a target and visual search with a target. For the study of browsing on Web pages, the users' scan paths usually present a rough and general pattern as an "F-shape" [7]. For the visual search with a target, the users' scan paths usually present the peripheral characteristic of visual search [8,9]. Animation prolonged the visual search time [10], and floating ads didn't impact the mode of visual search [9], however, it would just make users feel bored [11]. The studies about the effect of information locations on visual search demonstrated that, the search time was the longest when the information was located on the right side or lower side of the Web page [12,13], the upper information was found first, and the lower information was always neglected [14]. The previous studies about visual search focused on the information forms or locations, however, the study on the relation between them for the effect on users' visual search has not been reported yet.

Web pages are an important human-computer interface between people and the Internet. The studies about the human-computer interface have been gradually transformed from the machine-oriented to people-oriented. Thus, the philosophy for the Web page design has gradually been changed from that users have to adapt to Web pages to that the Web pages are designed according to users' habit. That is because the Web page design based on the habit of users' browsing can attract much more users' attention.

Eye-tracker can capture and record the detailed data of a user's eye movement, such as fixation duration and fixation count, which can reflect the human mental activities effectively. So far, eye-tracking has become an effective method to investigate the users' visual search behavior on Web pages [8] - [14].

This study focuses on exploring the relation between information forms and locations, as well as the floating ads for the impact of a Web page layout on visual search using eye-tracking. The results may be helpful for the Web page designers to find out users' interested part of Web pages and arrange the layout of different information forms on a Web page.

2 Experiments

2.1 Experiment 1: Relation between Information Forms and Locations

Participants. The participants were 50 undergraduates and postgraduates with the age range of 21 ~ 25 years old ($M = 23.0$, $SD = 1.3$), and half of them were female. All participants were right-handed, had normal or corrected-to-normal vision, often surf on the Internet and were skilled users of the mouse. None of them had the experience on eye movement experiment.

Apparatus. In the study, eye movements were recorded using the Tobii T120 eye-tracker made in Sweden at the rate of 120 HZ. The software of Tobii presented the Web pages automatically and recorded the data during participants' visual search on Web pages. The Web pages were displayed on the screen which

is a 19 LCD monitor with resolution set to 1024×768 pixels and at the refresh rate of 60 HZ. The distance between participant and screen was about 60 cm.

Experimental Material and Design. Studies about the layout of a Web page usually divide the Web page into several parts and the divisions were different. In some researches, the Web page was divided into upper and lower or left and right roughly; some divisions were in according with the four quadrants; and some were divided into 9 locations, etc. Actually, the layout of a Web page usually likes a tortoise shell that the center area is added on the base of 4 quadrants and each location is about 20 percent of the Web page, as shown in Fig. 1. In this study, this method is used for the layout of Web pages. The forms of the information are text and picture which are commonly used on the Web page. Text is the Chinese phrase consisted of 3 ~ 6 Chinese words, and picture is a well-known logo picture, such as Sony or Motorola.

In order to investigate the effect of the layout of 5 different locations (center and 4 quadrants) and the forms of information (text and picture) on users' visual search efficiency, we designed two Web pages on every location, thus the total number of Web pages was 10. The topics of the 10 Web pages were common to people involving mobile, diet, clothing and so on. In the two Web pages of each location, the information form on one page was text, and that on the other page was a logo picture. Each search target was presented only once on every Web page. In our study, all participants didn't know the target forms, and the target was described by the text on the pre-page. Moreover, all the participants were randomly divided into 10 groups.



Fig. 1. Five locations defined on a Web page. The information form on each location is text or a logo picture, and participants don't know the information form before the experiment.

Procedure. At the beginning of the experiment, participants were given the target description on the pre-page (whether the target was text or picture, the target was described as the text, and all participants didn't know the information form); then participants should find out the target on the search-page and if they find it, they would click the left mouse button on the target. After that, participants could have a rest and then they could click into the next task. The eye-tracker recorded the whole process of participants' visual search.

2.2 Experiment 2: Impact of the Floating Ads on Visual Search

Participants. The participants were 50 undergraduates and postgraduates with the age range of 18 ~ 27 years old ($M = 23.0$, $SD = 2.0$), and 26 were female. All participants were right-handed, had normal or corrected-to-normal vision, often surf on the Internet and were skilled users of the mouse. None of them had the experience on eye movement experiment.

Apparatus. The apparatus were the same as Experiment 1.

Experimental Material and Design. Floating ads were added on the base of the experimental material in Experiment 1. The size and floating speed of the floating ads were the same as the real situation, the pathway of the floating ads were different on each Web page and the floating ads appeared on random position. And others of the design were the same as Experiment 1.

Procedure. The procedure was the same as Experiment 1.

3 Results

In the study, we analyzed fixation duration during participants searching text or logo picture which was arranged on each location including the center area, quadrant I, II, III and IV on Web pages.

The fixation duration is the sum of all the fixations length while completing a visual search task, and the shorter the fixation duration is, the higher the visual search efficiency would be. Eye-movement researchers have established 100 ms as the minimum amount of time necessary for a pause to be considered a fixation [15] - [17]. Therefore, this study based on the measures that the fixation duration with a minimum threshold of 100 ms.

3.1 Relation between Information Forms and Locations

By analyzing the fixation duration, there was a significant main effect on information forms [$F(1,490) = 76.64$, $P < 0.001$], and also a significant main effect on locations [$F(4,490) = 8.34$, $P < 0.001$]. There was also a significant interaction between information forms and locations [$F(4,490) = 11.16$, $P < 0.001$].

Fixation Duration in Text vs. Logo Picture on Each Location. As shown in Fig. 2, there was a significant difference between text and logo picture on the fixation duration when the target was on each location. Except the quadrant III, the fixation duration in text was significantly longer than logo picture, which indicates that picture has superiority to text on all the locations except on quadrant III.

Fixation Duration in Text on Each Location. Figure 3 shows the fixation duration in text on each location. The fixation duration in the center area or quadrant IV was significantly longer than in other quadrants. However, there was no significant difference between the center area and quadrant IV. And there was also no significant difference between every two among quadrant I, II and III. Therefore, the results show that it is the worst to arrange text in the center area or quadrant IV.

Fixation Duration in Logo Picture on Each Location. Figure 4 shows the fixation duration in logo picture on each location. The fixation duration in quadrant II was significantly shorter than in other locations. However, the quadrant III was significantly longer than other locations. The results indicate that it is the worst to arrange the logo picture in quadrant III.

3.2 Effect of Floating Ads on Locations

To investigate the effect of floating ads on locations, we analyzed the fixation duration between the Web page with floating ads and without floating ads on each location. No matter what location the information is placed on the Web page, there is no significant difference between with floating ads and without floating ads on the fixation duration, which indicates that floating ads has no significant effect on the locations (see Table 1).

4 Discussion

4.1 Relation between Information Forms and Locations

Glaser WR. and Glaser MO. reported that picture had superiority effect [18], which had been demonstrated in early research [19]. When users viewing picture and text which have a certain relation in content, they usually have a glance at picture first, then read text, and view picture again. This pattern has been reported for viewers looking at magazine advertisements containing text and picture [20]. In our study, the fixation duration on logo picture is significantly shorter than text on locations but quadrant III (see Fig. 2). The result also indicates that picture has superiority effect to text.

Previous studies showed that visual search behavior sometimes would be guided by the physics location in display [21]. Though picture has superiority effect to text, Fig. 2 shows clearly that when information was arranged in

quadrant III, fixation duration in logo picture was significantly longer than text, which was opposite to other locations. Furthermore, Fig. 3 and Fig. 4 show that for the text, the fixation duration in quadrant IV and the center area was the longest; in contrast, for the picture, that is in quadrant III.

Early research indicated that eyes need time to extract information, and long fixation duration was correlated with extracting more information [22]. Moreover, the difficulty of extracting information also may have an effect on fixation duration. Therefore, hardly identify or dense information display need long fixation duration [23]. In the study, we considered that there was a correlation between the fixation duration and the difficulty of users' visual search. The fixation duration would be shorter if it is easier to find out the information, otherwise the fixation duration would be longer. A website usually contains a series of Web pages, which were linked by Web navigation, to place different information. The result shows that the fixation duration in quadrant II is shorter whether the information is text or picture, which suggests that important links, such as navigation, should be placed in quadrant II. It is easier and more convenient for users to find out other related Web pages and view their interested information. The results also show that the fixation duration in text placed in quadrant IV and the center area are longer, therefore it is best not to place important text information in quadrant IV and the center area, so as not to increase difficulty for users' visual search. In contrast to picture, that is the quadrant III, so it is best not to place important picture information in quadrant III, which would be advantage for users' visual search.

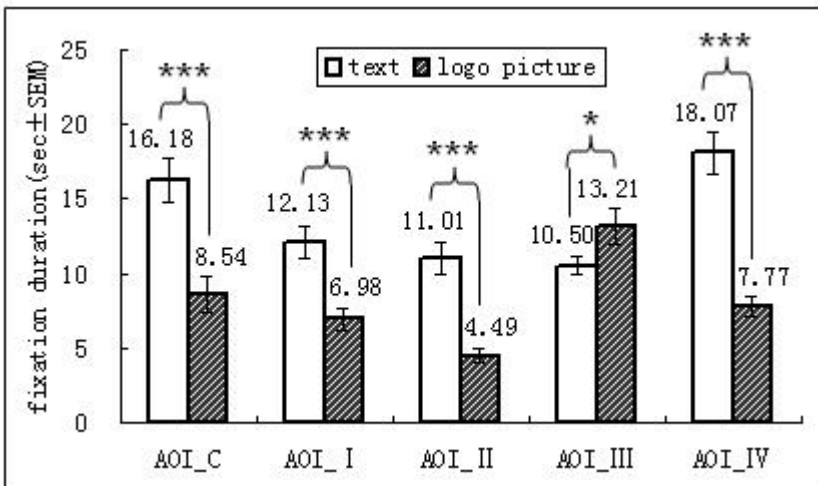


Fig. 2. The fixation duration when the text and logo picture information were on 5 locations. The fixation duration in logo picture was significantly shorter than text on all the locations except on quadrant III (*: the significant level of F-check < 0.05; ***: < 0.001).

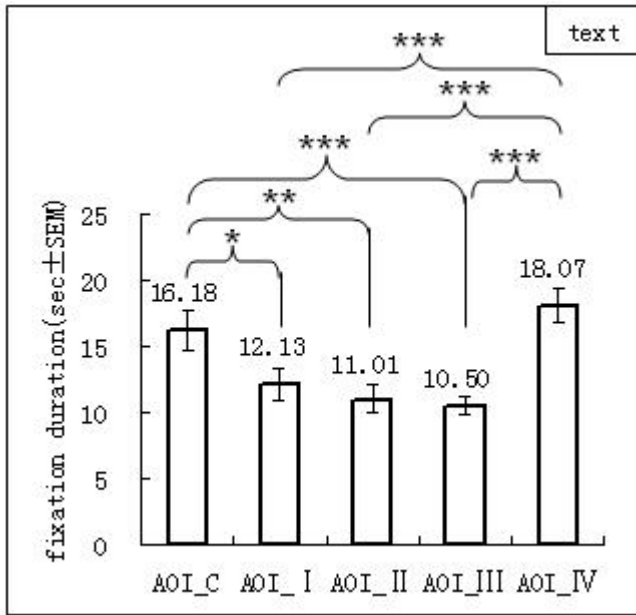


Fig. 3. The fixation duration in text on each location. The fixation duration in the center area or quadrant IV is significantly longer than in quadrant I, II and III (*: the significant level of F-check < 0.05; **: < 0.01; ***: < 0.001).

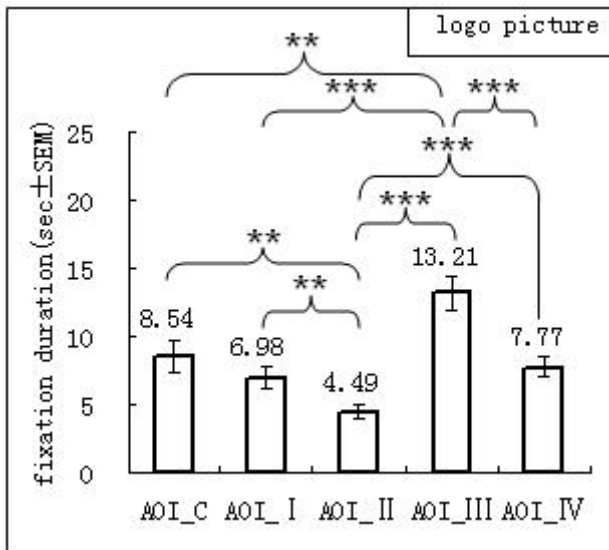


Fig. 4. The fixation duration in logo picture on each location. The fixation duration in quadrant II is the shortest and in quadrant III is the longest (**: the significant level of F-check < 0.01; ***: < 0.001).

Table 1. The fixation duration of each location on the Web page without floating ads and with floating ads ($M \pm SEM$), there was no significant difference between with floating ads and without floating ads

Location	Without floating ads(s)	With floating ads(s)	Significant level
AOI_C	12.01 $\pm$ 1.70	11.57 $\pm$ 1.64	$F(1,98) = 0.10, P = 0.75$
AOI_I	9.80 $\pm$ 1.39	9.70 $\pm$ 1.37	$F(1,98) = 0.01, P = 0.92$
AOI_II	7.82 $\pm$ 1.11	7.95 $\pm$ 1.12	$F(1,98) = 0.02, P = 0.89$
AOI_III	11.76 $\pm$ 1.66	14.11 $\pm$ 2.00	$F(1,98) = 3.87, P = 0.05$
AOI_IV	12.94 $\pm$ 1.83	11.40 $\pm$ 1.61	$F(1,98) = 1.70, P = 0.20$

Through analyzing the first fixation, we found that 90% of the participants' first fixation was in the center area. The result shows that the central area is the first location that participants run their eyes over the Web page. According to the inhibition of return (IOR) theory [24], participants' attention returned to the center area was inhibited. Therefore, participants' visual search in this area needs to spend more time, and it is not easy to find out the target quickly. The study suggests that when text is placed in the center area, it is best to use larger size or bright-colored title which can distinguish the main text to help users to search their desired information quickly.

4.2 Impact of Floating Ads on Each Location

As shown in Table 1, there is no significant difference on fixation duration between without floating ads and with floating ads when the target was on each location, which also demonstrated the recent studies [9,11] that users often consider that the floating ads have no relation to their desired information, and don't pay more attention to them, thus the floating ads have no significant impact on visual search efficiency.

5 Conclusion

This study investigated the relation between information forms and locations, as well as the floating ads for the impact of a Web page layout on visual search. Through analyzing the fixation duration, the results show that for the text, the quadrant IV (right lower) and the center area are longer than other locations; for the picture, quadrant III (left lower) is the longest. Whether the target is text or picture, quadrant II (left upper) is shorter. And no matter what location the information is placed on the Web page, there is no significant difference between the page with floating ads and without floating ads. The results indicate that the important information whether text or picture is suitable to place in quadrant II; however, it is the worst to arrange text information in quadrant IV and the center area or picture information in quadrant III on the layout of Web pages.

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